CompTIA Network+ (N10-009)

Chapter 5: Compare and Contrast Transmission Media & Transceivers

A Study Guide for Aspiring Network Professionals

By Hassen Hannachi | N10-009 Study Series | April 2026

Chapter Overview

Chapter 5 of CompTIA Network+ N10-009 focuses on the physical layer of networking — the cables, wireless standards, cellular technologies, connectors, and transceivers that carry data between devices. Understanding these topics is essential both for the exam and for day-to-day work in any network environment.

This chapter covers four major exam objective areas:

- Wireless — 802.11 standards and Wi-Fi generations
- Cellular — LTE, NB-IoT, GSM, and satellite
- Wired — Fibre, copper, coaxial, and cabling standards
- Transceivers & Connector Types — Form factors, protocols, and physical connectors

Section 1: Wireless

Wireless networking transmits data using radio frequency (RF) signals through the air rather than physical cables. The IEEE 802.11 family of standards defines how Wi-Fi networks operate, evolving over decades to deliver faster speeds and more reliable connections.

802.11 Standards

The IEEE (Institute of Electrical and Electronics Engineers) publishes all 802.11 wireless standards. Each generation improves on speed, frequency bands, range, and capacity.

Standard	Wi-Fi Name	Frequency	Max Speed	Key Feature
802.11b	Wi-Fi 1	2.4 GHz	11 Mbps	First widely adopted standard (1999)
802.11a	Wi-Fi 2	5 GHz	54 Mbps	Less interference, shorter range
802.11g	Wi-Fi 3	2.4 GHz	54 Mbps	Backward-compatible with 802.11b
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps	MIMO introduced; dual-band
802.11ac	Wi-Fi 5	5 GHz	3.5+ Gbps	MU-MIMO; beamforming; wide channels
802.11ax	Wi-Fi 6/6E	2.4 / 5 / 6 GHz	9.6 Gbps	OFDMA, BSS Coloring, dense environments
802.11be	Wi-Fi 7	2.4 / 5 / 6 GHz	46 Gbps	Multi-Link Operation (MLO); EHT

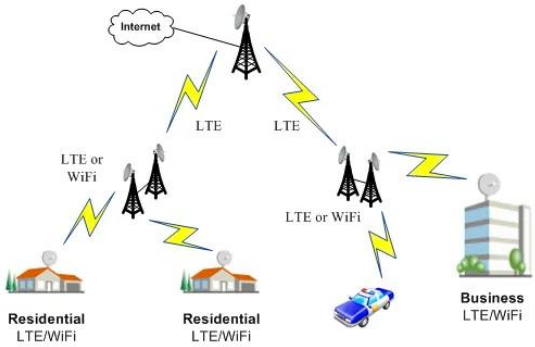
Key 802.11 Concepts to Know

- MIMO (Multiple Input Multiple Output): Uses multiple antennas to send/receive simultaneously, boosting throughput — introduced in 802.11n.
- MU-MIMO (Multi-User MIMO): Allows the AP to communicate with multiple clients at once (802.11ac/ax).
- OFDMA (Orthogonal Frequency Division Multiple Access): Splits channels into sub-channels for more efficient delivery in dense environments — key feature of Wi-Fi 6.
- Beamforming: Focuses the wireless signal toward specific clients rather than broadcasting in all directions, improving range and speed.
- BSS Coloring: Reduces interference in dense deployments by tagging frames with a color identifier so devices can ignore signals from neighboring networks.
- Channels: 2.4 GHz has 11 channels (3 non-overlapping: 1, 6, 11); 5 GHz has many more non-overlapping channels — reducing co-channel interference.

> **Exam Tip:** Know the frequency bands for each standard. 802.11b and g are 2.4 GHz only; 802.11a is 5 GHz only; 802.11n, ac, and ax are dual or tri-band. 802.11ax (Wi-Fi 6E) adds the 6 GHz band.

Section 2: Cellular

Cellular networks use radio towers divided into geographic cells to provide wide-area wireless coverage. Unlike Wi-Fi, cellular can span cities, countries, and continents. The exam tests knowledge of four cellular technologies.



Long Term Evolution (LTE)

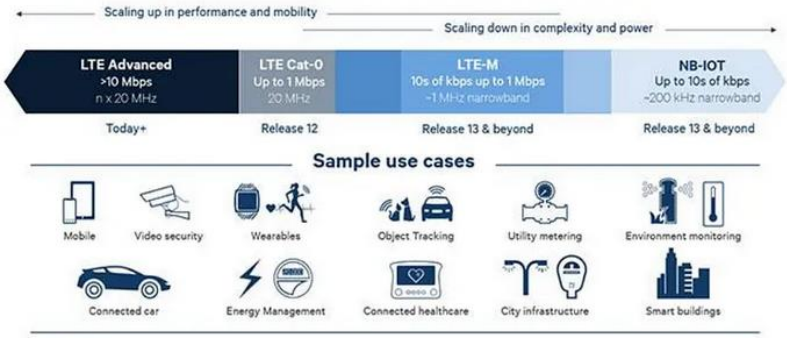
LTE — often marketed as 4G LTE — is the most widely deployed cellular standard globally. It represents a significant leap over previous 3G technologies and forms the backbone of modern mobile data networks.

- Theoretical speeds: Up to 300 Mbps download / 75 Mbps upload (real-world is much less).
- LTE-Advanced (LTE-A): Extends LTE with carrier aggregation, reaching speeds over 1 Gbps.
- LTE-M (LTE for Machines): A low-power variant designed for IoT devices; supports voice and mobility.
- Uses licensed spectrum managed by mobile carriers (e.g., AT&T, Verizon, T-Mobile).
- Packet-switched: All data, including voice (VoLTE), is transmitted as IP packets.

> **Real World:** When you connect your phone to the internet away from Wi-Fi, you are almost certainly using LTE or 5G NR (New Radio). LTE is the global standard that 5G networks also rely on as a fallback.

Narrowband-IoT (NB-IoT)

NB-IoT is a low-power wide-area (LPWA) technology standardized by 3GPP to connect IoT devices that send small amounts of data infrequently.



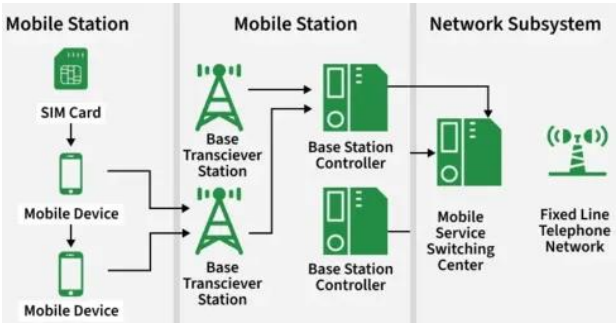
- Operates in licensed spectrum — reliable and interference-free compared to unlicensed alternatives.
- Designed for deep indoor penetration (basements, underground meters).

- Supports massive device density — thousands of devices per cell.
- Very low power consumption — devices can run on small batteries for years.
- Low data rates (up to ~250 kbps): suitable for sensors, smart meters, asset trackers.
- Not designed for high mobility (phones moving at speed); suited for stationary or slow-moving devices.

> **Exam Tip:** NB-IoT competes with LoRaWAN and Sigfox for IoT connectivity. Know that NB-IoT uses licensed spectrum (more reliable) while LoRaWAN uses unlicensed spectrum (cheaper, less guaranteed).

Global System for Mobile Communication (GSM)

GSM is the foundational 2G cellular standard, first deployed in 1991 and still in use as a fallback in many parts of the world. It introduced digital mobile communication and the SIM card.

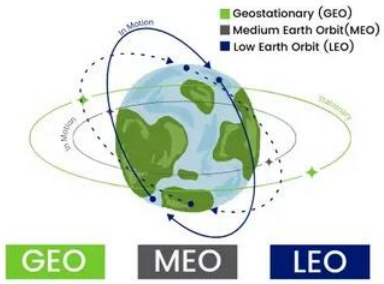


- Frequency bands: 850 MHz, 900 MHz, 1800 MHz, 1900 MHz.
- TDMA (Time Division Multiple Access): Multiple users share a frequency channel by dividing it into time slots.
- Introduced the SIM (Subscriber Identity Module) card — a removable card identifying the subscriber.
- Data extensions: GPRS (2.5G) and EDGE (2.75G) added basic mobile data.
- Still relevant for IoT devices using GSM/GPRS modules in regions without LTE coverage.

> **Exam Tip:** GSM is important context for the exam. Know that CDMA (used by Verizon historically) is the main alternative to GSM. Most modern networks have moved to LTE and 5G, but understanding GSM helps with troubleshooting legacy devices.

Satellite

Satellite internet provides connectivity via communication satellites in Earth orbit. It is the go-to solution for remote areas without terrestrial infrastructure.



Orbit Type	Altitude	Latency	Example
GEO (Geostationary)	~35,786 km	500–700 ms	HughesNet, ViaSat
MEO (Medium Earth Orbit)	~2,000–35,786 km	~100–200 ms	O3b (SES)
LEO (Low Earth Orbit)	~550–1,200 km	20–40 ms	Starlink, OneWeb

- GEO satellites orbit at a fixed point above the equator — the dish can be fixed but latency is very high.
- LEO satellites (like Starlink) orbit much closer, offering dramatically lower latency but require tracking antennas as satellites move across the sky.
- Use cases: remote locations, maritime, aviation, disaster recovery, military.

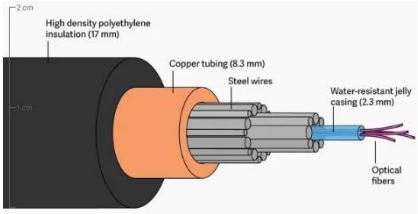
> **Key Point:** Satellite's biggest challenge is latency. GEO latency (~600ms round trip) makes real-time applications (VoIP, gaming) difficult. LEO systems like Starlink aim to solve this with ~30–40ms latency.

Section 3: Wired Media

Wired transmission media remain the backbone of enterprise networking. Physical cables offer predictable performance, security, and high throughput compared to wireless alternatives.

Fibre Ethernet and OM Standards

Fibre optic cables transmit data as pulses of light through a glass or plastic core. They are immune to electromagnetic interference (EMI), support very long distances, and carry massive bandwidth.



Multimode Fibre (MMF) — OM Standards

Multimode fibre has a larger core (50 or 62.5 microns) and allows multiple light modes to travel simultaneously. Used for shorter distances, typically within a data center or campus.

Standard	Core Size	Max Distance (10G)	Max Distance (100G)	Color
OM1	62.5 µm	33 m	N/A	Orange
OM2	50 µm	82 m	N/A	Orange
OM3	50 µm	300 m	100 m	Aqua
OM4	50 µm	400 m	150 m	Aqua / Violet
OM5	50 µm	400 m	150 m (WDM)	Lime Green

> **Memory Aid:** OM1 and OM2 are legacy (orange). OM3 and OM4 are aqua — the modern standards. OM5 is lime green and designed for Wideband Multimode Fibre (WBMMF) using multiple wavelengths.

Single-Mode vs Multimode Fibre

Property	Single-Mode (SMF)	Multimode (MMF)
Core Diameter	8–10 µm (very narrow)	50 or 62.5 µm (wider)
Light Source	Laser (single wavelength)	LED or VCSEL
Distance	Up to 100 km+	Up to 550 m (OM3/OM4)
Bandwidth	Extremely high	High but limited by modal dispersion
Cost	Higher (laser equipment)	Lower (LED equipment)
Use Case	Long-haul, WAN, ISP	Data centers, campus, LAN
Cable Color	Yellow (OS1/OS2)	Orange, Aqua, or Green (OM1–OM5)

> **Exam Tip:** Single-mode = long distance, laser, yellow jacket. Multimode = short distance, LED, color-coded by OM standard. The key differentiator is core size and distance.

Direct Attach Copper (DAC) Cables

DAC cables are passive or active copper twinaxial cables with transceivers pre-attached at both ends. They are a cost-effective alternative to fibre for short-distance, high-speed connections within racks or between adjacent racks.

- Typical lengths: 1–7 meters (passive); up to 15 meters (active DAC).
- Common speeds: 10G, 25G, 40G, 100G, 400G.
- Used for server-to-switch and switch-to-switch connections inside data centers.
- No optical conversion needed — reduces latency and cost vs. fibre + transceiver.
- Form factors: SFP+, QSFP+, QSFP28 ends (pre-installed).

> **Key Point:** DAC cables are the cheapest and lowest-latency option for short data center runs. They cannot be used for long distances — that is where fibre takes over.

Coaxial Cable

Coaxial (coax) cable consists of a central copper conductor, surrounded by insulation, a metallic shield, and an outer jacket. The shield protects against EMI and allows the cable to carry high-frequency signals.

- RG-6: Standard for cable TV, satellite TV, and broadband internet to the home.
- RG-59: Older standard; lower quality, used for short CCTV runs and older cable TV.
- RG-11: Thicker, lower attenuation; used for longer cable runs.
- Characteristic impedance: 75 ohms for video/broadband; 50 ohms for RF/networking.
- Still widely used for cable internet (DOCSIS), CCTV cameras, and antenna connections.

Cable Speed

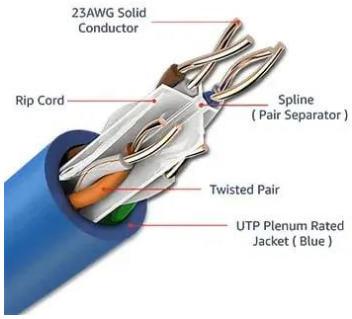
Cable speed refers to the rated maximum data transfer rate of a cable type. Key factors affecting speed include:

- Category (Cat) rating for copper UTP: Cat5e supports 1 Gbps; Cat6 supports 10 Gbps (short runs); Cat6a supports 10 Gbps at 100m; Cat7/8 for higher speeds.
- Fibre type and OS/OM rating define supported speeds and distances.
- Coaxial speed depends on DOCSIS version — DOCSIS 3.1 supports multi-Gbps.
- DAC cable speed is defined by the transceiver standard (e.g., 25GBASE-CR for 25G DAC).

> **Exam Tip:** For copper Ethernet: Cat5e = 1G, Cat6 = 10G (short distances), Cat6a = 10G (100m). These are commonly tested values.

Plenum vs Non-Plenum Cable

Plenum is a type of jacket in Ethernet cables. The major difference between plenum cables and other cables are ***jacket material***. In the plenum-rated cables use a ***fire-retardant*** material which burns slowly and emits ***less intoxicated*** fumes as compared to the non-plenum-rated cables.



Property	Plenum (CMP)	Non-Plenum / Riser (CMR) / PVC
Jacket Material	Low-smoke, fire-retardant (FEP/FHPE)	Standard PVC
Fire Behavior	Self-extinguishing; low toxic smoke	Burns more readily; emits toxic smoke
Installation Area	Air ducts, false ceilings, HVAC spaces	Walls, conduits, general runs
Cost	Higher (30–50% more)	Lower
NEC Requirement	Required in plenum spaces	Prohibited in plenum spaces

> **Important:** Plenum spaces are the air-handling spaces in buildings (above drop ceilings, under raised floors) where HVAC air circulates. Fire codes (NEC) require plenum-rated cable in these spaces because toxic smoke from burning PVC could circulate throughout the building.

Section 4: Transceivers

A transceiver is a device that can both transmit and receive signals. In networking, transceivers are modular components (often called pluggable optics) that convert electrical signals to optical (light) signals and vice versa, allowing network equipment to support different cable types and speeds.

Transceiver Protocols

The protocol defines the data rate and physical standard the transceiver implements:

Protocol	Speed	Media	Use Case
1000BASE-SX	1 Gbps	Multimode Fibre	Short-range LAN (up to 550m)
1000BASE-LX	1 Gbps	Single or Multimode	Medium range (up to 10 km SMF)
10GBASE-SR	10 Gbps	Multimode Fibre	Data center short reach
10GBASE-LR	10 Gbps	Single-Mode Fibre	Long reach (up to 10 km)
25GBASE-SR	25 Gbps	Multimode Fibre	Modern server uplinks
40GBASE-SR4	40 Gbps	Multimode Fibre (MPO)	High-speed data center links
100GBASE-LR4	100 Gbps	Single-Mode Fibre	Long-range 100G links
BiDi (Bidirectional)	1–100 Gbps	Single Fibre (SMF/MMF)	Transmit + receive on one strand using WDM

> **Key Concept:** SR = Short Range (multimode). LR = Long Range (single-mode). ER = Extended Range. ZR = Ultra-long range. These suffixes appear on transceivers and mean the same thing across speed tiers.

Transceiver Form Factors

Form factors define the physical shape and interface of the transceiver module. Network equipment (switches, routers, NICs) has slots that accept specific form factors.

Form Factor	Speed	Description
SFP (Small Form-factor Pluggable)	Up to 4 Gbps	Original standard; supports 1G Ethernet, Fibre Channel
SFP+ (Enhanced SFP)	Up to 16 Gbps	Supports 10G Ethernet; most common in enterprise
SFP28	Up to 28 Gbps	Supports 25G; used in modern data centers

Form Factor	Speed	Description
QSFP (Quad SFP)	4x10 Gbps = 40 Gbps	Four channels; supports 40G
QSFP+ (Enhanced QSFP)	Up to 41 Gbps	Common for 40G links
QSFP28	4x25 Gbps = 100 Gbps	Most common 100G form factor
QSFP-DD (Double Density)	8x25G or 8x50G	400G; double the lanes of QSFP28
CFP / CFP2 / CFP4	100G+	Larger form factors for high-power optical modules

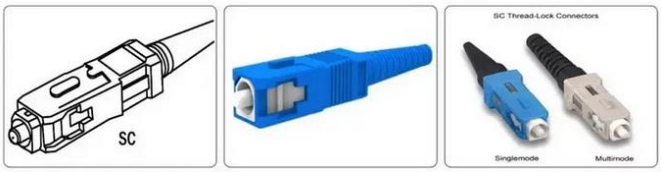
> **Exam Tip:** SFP = 1G; SFP+ = 10G; QSFP+ = 40G; QSFP28 = 100G. These are the four most tested transceiver form factors on the Network+ exam. Also remember that transceivers are hot-swappable — you can replace them without powering down the switch.

Section 5: Connector Types

Connectors are the physical interfaces at the end of cables. Knowing which connector goes with which cable type and use case is essential for the Network+ exam and for hands-on network work.

Fibre Optic Connectors

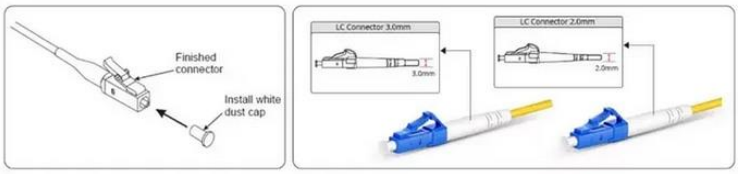
SC (Subscriber Connector / Standard Connector)



- Type: Fibre optic (single-mode and multimode)
- Shape: Square, push-pull latching mechanism — click in, pull out.
- Core size: 2.5mm ferrule.
- Use case: Older enterprise networking, broadband, cable TV (FTTX).
- Duplex SC: Two SC connectors side by side for TX and RX.

> **Memory Aid:** SC = Square Connector. Its square shape and snap-in mechanism make it easy to identify.

LC (Lucent Connector / Little Connector)



- Type: Fibre optic (single-mode and multimode)
- Shape: Small form factor — half the size of SC; uses a 1.25mm ferrule.
- Latching: Tab latch (similar to an RJ45 clip).
- Use case: Most common fibre connector in modern enterprise and data center environments.
- Duplex LC: Two LC connectors in a single housing — the standard for SFP transceivers.

> **Memory Aid:** LC = Little Connector. It is the small, modern fibre connector you will see most often in data centers.

ST (Straight Tip)

- Type: Fibre optic (multimode, occasionally single-mode)
- Shape: Cylindrical with a bayonet-style twist-lock mechanism.
- Core size: 2.5mm ferrule.

- Use case: Older fibre installations, campus networks, military applications.
- Less common in new installations — SC and LC have largely replaced it.

> **Memory Aid:** ST = Stab and Twist. You insert and twist the bayonet lock to secure the connector.

Copper / Telephony Connectors

RJ11

- Type: Copper telephone connector
- Pins: 6 positions, 2 or 4 conductors (6P2C or 6P4C)
- Use case: Analog telephone lines (POTS), DSL modems, home phone wiring.
- Smaller than RJ45; will not fit in an RJ45 jack correctly.

> **Key Difference:** RJ11 is for telephone/DSL (narrower, 6-position). RJ45 is for Ethernet (wider, 8-position). Knowing the difference is a basic but commonly tested Network+ question.

RJ45

- Type: Copper Ethernet connector
- Pins: 8 positions, 8 conductors (8P8C)
- Use case: Ethernet cabling (Cat5e, Cat6, Cat6a) for LAN connections.
- Wiring standards: T568A and T568B define which wire pair goes in which position.
- Crossover cable: TX and RX pairs swapped — used for direct device-to-device connections (legacy; modern devices use Auto-MDIX).

> **Exam Tip:** RJ45 with T568B wiring is the most common Ethernet cabling standard in North America. Know the color order for both T568A and T568B for the exam.

Coaxial Connectors

F-Type Connector

- Type: Coaxial (RG-6, RG-59)
- Mechanism: Threaded screw-on (male/female).
- Use case: Cable TV, satellite TV, broadband cable internet (DOCSIS modem connections), over-the-air antenna.
- Impedance: 75 ohms.
- The connector that screws onto the back of your cable modem or TV.

> **Memory Aid:** F-type = the connector on your cable TV or cable modem. It screws on.

BNC (Bayonet Neill-Concelman)

BNC Male Connector

BNC Female Connector

- Type: Coaxial
- Mechanism: Bayonet-style twist-lock (quarter-turn to lock/unlock).
- Use case: 10BASE2 (Thinnet) legacy Ethernet, professional video (SDI), test equipment, CCTV.
- Impedance: 50 ohms (networking) or 75 ohms (video).
- Largely legacy in networking but still common in broadcast video and security cameras.

BNC T-Connector

- Type: Coaxial adapter
- Shape: T-shaped; has one BNC female and two BNC male connectors.
- Use case: 10BASE2 (Thinnet) Ethernet bus topology — connected each network device to the coax backbone.
- The device's NIC connected to the middle of the T; the coax segments connected to each end.

> **History Note:** BNC T-connectors were essential in early Ethernet bus networks (10BASE2). If one T-connector or terminator failed, the entire segment went down — a key limitation of bus topology.

BNC Barrel Connector

- Type: Coaxial adapter
- Shape: Cylindrical; joins two BNC male connectors end-to-end.
- Use case: Extending a coaxial cable run by joining two cable segments.
- Used in 10BASE2 networks and CCTV/video installations.
- Each barrel connector introduces signal loss (attenuation) — should be minimized.

Quick Reference: Connector Summary

Connector	Type	Media	Use Case	Key Feature
SC	Fibre	SMF / MMF	Older enterprise, FTTX	Square, push-pull, 2.5mm
LC	Fibre	SMF / MMF	Modern data center (SFP)	Small, tab-latch, 1.25mm
ST	Fibre	MMF (legacy)	Campus / military / legacy	Bayonet twist-lock
RJ11	Copper	2-4 conductor	Telephone / DSL	6P2C/4C, narrow
RJ45	Copper	8 conductor	Ethernet LAN	8P8C, T568A/B wiring
F-Type	Coaxial	RG-6 / RG-59	Cable TV / broadband modem	Threaded screw-on
BNC	Coaxial	RG-58	Legacy 10BASE2 / video	Bayonet quarter-turn
BNC T	Coaxial adapter	RG-58	10BASE2 bus topology	Joins NIC to segment
BNC Barrel	Coaxial adapter	RG-58	Extending coax runs	Joins two cables end-to-end

Chapter 5 — Exam Tips & Key Takeaways

Wireless

- Know all 802.11 standards by frequency and max speed — especially 802.11n (dual-band, MIMO), 802.11ac (5 GHz, MU-MIMO), and 802.11ax (OFDMA, Wi-Fi 6, 6 GHz added).
- 2.4 GHz = longer range, more interference (microwaves, Bluetooth). 5 GHz = shorter range, less interference, more channels.

Cellular

- LTE = 4G mobile broadband. NB-IoT = low-power IoT in licensed spectrum. GSM = 2G foundation with SIM cards. Satellite = remote areas with GEO high latency vs. LEO low latency.

Wired

- Single-mode fibre = yellow jacket, laser, long distance. Multimode = colored (aqua/orange/green), LED, short distance.
- OM1/OM2 are orange (legacy); OM3/OM4 are aqua; OM5 is lime green.
- Plenum cable is required where HVAC air circulates — it produces less toxic smoke when burning.
- DAC cables: cheap, low-latency, copper twinax for rack-to-rack runs in data centers.

Transceivers

- SFP = 1G, SFP+ = 10G, QSFP+ = 40G, QSFP28 = 100G. These are hot-swappable.
- SR = short range (multimode), LR = long range (single-mode).

Connectors

- SC = square, push-pull fibre. LC = little/small, tab-latch fibre (modern standard). ST = stab and twist, bayonet lock (legacy).
- RJ11 = phone (6 pins, narrow). RJ45 = Ethernet (8 pins, wider).
- F-type = cable TV/modem, threaded. BNC = legacy coax, bayonet. BNC T = bus topology. BNC Barrel = extends coax.

> **Final Tip:** For the Network+ exam, focus on what differentiates similar options: SC vs LC (size), single-mode vs multimode (core size, distance, color), RJ11 vs RJ45 (pins, use case), plenum vs non-plenum (fire rating, air space requirement). These contrasts are exactly what exam questions test.

Good luck on your Network+ N10-009 exam! *You've got this.*